

Jacob Mattie

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Distributed systems specialist: integrating embedded control systems, Linux infrastructure, and telemetry pipelines into automated operational workflows.

Programming

Bash Scripting Python
C++ SQL
HTML5 / SCSS MATLAB

Connectivity

Ethernet REST APIs
HTTP WiFi
JSON UART

Middleware

EPICS systemd
Gunicorn Linux
journalctl Redis

— BSc. Mathematics, 2022, Simon Fraser University —

Work Experience

Investment Analyst — Data Architecture & Backend Systems

Aligned Capital Partners / River Wealth

Vancouver, BC
11/2024 - Ongoing

Q: What data exists that might let us predict stock growth and market behaviour?

- Engineered **data ingestion** workflows across multiple methods (HTTP requests, file-based feeds, web scraping), using stateful retries and validation checks to handle unreliable upstream sources.
- Fully provisioned and maintained headless **production Linux infrastructure** from bare-metal deployment through service orchestration (systemd, Nginx, Gunicorn).
- Built Python-based reporting systems that **automated data monitoring** workflows, reducing company-wide labour load by 20% and mitigating potential of human error.

Test Technologist

IBC Technologies

Burnaby, BC
6/2022 – 9/2023

Q: How can we safely optimize heat transfer from a boiler's combustion chamber?

- Automated hardware validation processes through IoT ESP32 microcontrollers and PLC systems, streaming test data into **cloud-connected repositories** for analysis and report generation.
- Led certification and **validation of hardware/software systems** for boilers under Canadian and U.S. regulations, triaging issues and coordinating cross-team resolution to support two product releases.
- Built an embedded **ESP32 control system** for an assembly station upgrade, adding a laser-based obstruction sensor as an override to technician controls, resulting in a 15% throughput increase with higher operator safety.
- Built high-throughput data analysis workflows in MATLAB for multi-million-point datasets, **reducing test validation time by 85%**.

IT Technician (Internship)

BC Cancer Research Centre (BCCRC)

Vancouver, BC
09/2018 – 01/2019

Q: How much digital freedom can we grant researchers without compromising network security?

- Managed user access controls and permissions within a segregated research network, enforcing data isolation and **cybersecurity** policies across multiple user groups.
- Managed, diagnosed, and resolved **TCP/IP-based** network connectivity issues across a segmented domain environment, including endpoint user access and inter-device communication.
- Assisted in network recovery during facility-wide outage caused by a malware incident, supporting administrators in restoring connectivity and validating **system integrity** across endpoints.

Volunteering

Automation Technologist

TRIUMF - Canada's Particle Accelerator

Vancouver, British Columbia
05/2025 - Ongoing

Q: What modern technological advances have resolved longstanding engineering problems?

- Contributed the **technical foundations** of 3 scientific publications (in preparation) on distributed control systems in nuclear physics laboratories.
- Delivered a containerized EPICS **deployment pipeline** using Bash, leading a modernization initiative for legacy middleware operating over \$1.5B in local infrastructure.
- Coded a Raspberry Pi-based stepper motor control system, merging hardware wiring, driver design, and Python interface through a transistor array for **automated and repeatable** experimental control.